

CoRoT-2b

R-0224

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240814 · created 2026-07-17T02:14:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460536.7266 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.057
Transit depth (Rp/Rs)^2	3.8e-05 +/- 0.007 [%]
Orbital Inclination (inc)	83.86 +/- 2.12
Airmass coefficient 1 (a1)	1.36 +/- 0.21
Airmass coefficient 2 (a2)	-0.23 +/- 0.32
Scatter in the residuals of the lightcurve fit is	15.05 %
Best Comparison Star	#1 - [531, 415]
Optimal Aperture	3.11
Optimal Annulus	16.79
Transit Duration (day)	0.056 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.057 vs 0.1667 (deviation 1.8σ)
Duration fit vs published	0.056 d vs 0.095 d (deviation 0.96σ)
Mid-time O-C	-1.5 min
Depth SNR	0.0
Residual scatter	15.05 %

Light curve

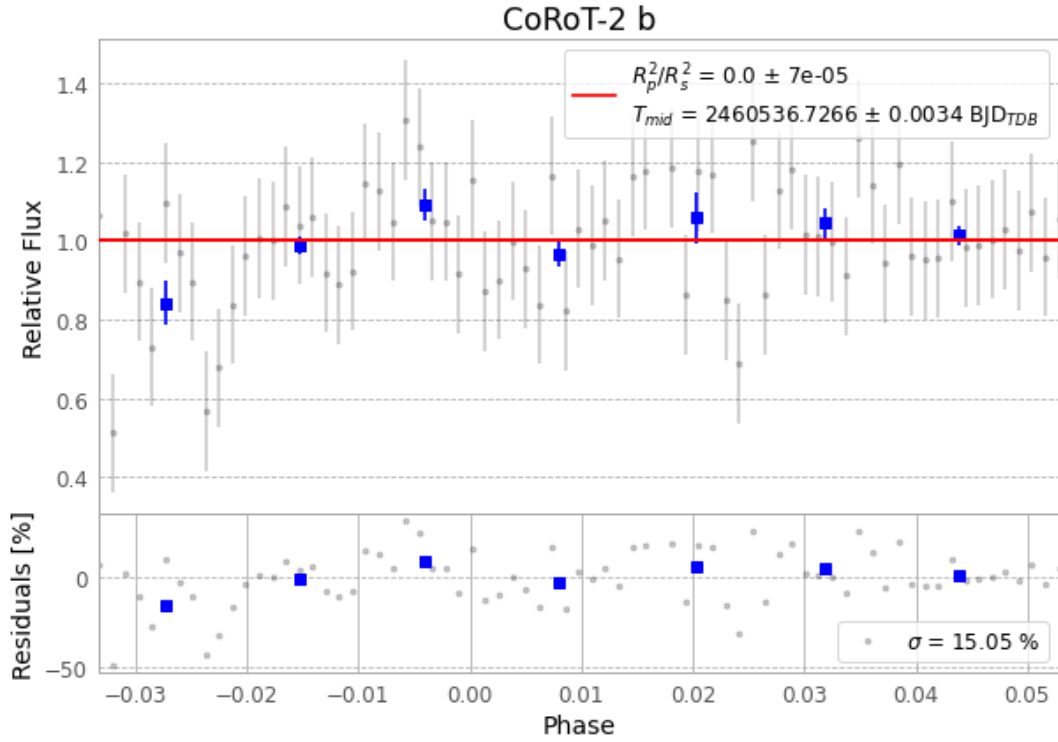


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-08-13.png (SHA-256 f1fdf94f8f4f1c98...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [121, 392], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e79e646d8caa7575...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), CoRoT-2240814035415.FITS → CoRoT-2240814073015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240814.log (817902613420...), FinalLightCurve_CoRoT-2 b_2024-08-13.png (f1fdf94f8f4f...), FinalLightCurve_CoRoT-2 b_2024-08-13.csv (9a0bde01ecb6...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 02:14Z.